

MATH 110C, HOMEWORK 3

DUE TUESDAY APRIL 30

Exercise 1 Let F be a field and $K = F(T)$ its field of rational fractions.

(a). Show that $X^n - T$ is an irreducible polynomial of $K[X]$ for all $n \geq 1$.

(b). Let L be an algebraic closure of K . Show that $[L : K]$ is infinite.

Exercise 2 Let $\alpha \in \mathbb{C}$. Assume α is algebraic over $\mathbb{Q}$ of degree $n > 1$.

Show that there exists c a positive real number such that given any $p \in \mathbb{Z}$ and $q \in \mathbb{Z}$, $q \neq 0$, we have $|\alpha - \frac{p}{q}| > \frac{c}{q^n}$.

(Hint: in the case $\frac{p}{q} \in [\alpha - 1, \alpha + 1]$, use the Mean Value Theorem for $P \in \mathbb{Z}[X]$ such that $P(\alpha) = 0$ and P is irreducible in $\mathbb{Q}[X]$).

Exercise 3 Let $(\alpha_k)_{k \geq 0}$ be a sequence of elements of $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ such that $a_k \neq 0$ for infinitely many k .

Let $\alpha = \sum_{k \geq 0} \frac{\alpha_k}{10^{k!}}$. Deduce from Exercise 2 that α is transcendental over $\mathbb{Q}$.

Exercise 4 Do Problem 2 in Exercises 51.5, p.311 and Problem 5 in Exercises 52.21, p.320 of the book.